

Kestrel 3.0 release notes

Released 30 September 2026 · [Download](#) · [Full changelog](#) · [Upgrade guide](#)

Highlights

Kestrel 3.0 is the first major release in two years. It replaces the scenario engine, halves tail latency under load, and makes every run reproducible from a single `kestrel.lock` file.

47% lower p99 latency in the reference scenario, from 356 ms to 188 ms.

Reproducible runs: seeds, versions, and targets are pinned in `kestrel.lock`.

Scenario files in TOML, checked by `kestrel check` before a run starts.

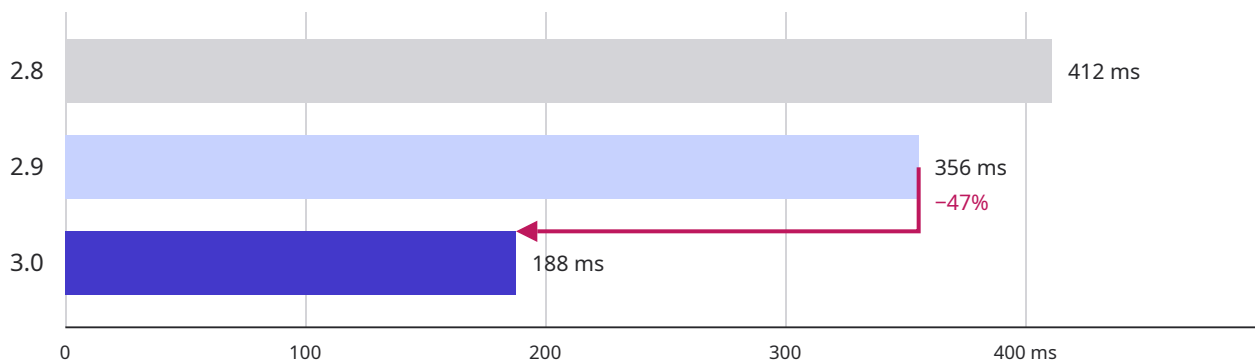


Figure 1. p99 latency in the reference scenario for 2.8, 2.9, and 3.0 (shorter is better)

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Breaking changes

Action required. YAML scenarios no longer load. Convert them before upgrading:

```
kestrel migrate scenarios/ --to toml --write
```

- Scenario files are TOML; `kestrel migrate` converts YAML files in place. ([#2210](#))
- The `--rps` flag is now `--rate` and accepts units such as `500/s`. ([#2187](#))
- Metrics are exported as OpenTelemetry by default; set `export = "statsd"` to keep the old format. ([#2143](#))



- macOS on Intel is no longer supported; the 2.9 LTS line remains available. ([#2231](#))

Added

- `kestrel.lock` pins the random seed, target versions, and plugin hashes for every run. ([#2102](#))
- `kestrel check` validates scenarios, including unreachable steps and unused variables. ([#2125](#))
- New open-loop arrival models:
 - `poisson` for independent arrivals; ([#2166](#))
 - `ramp` and `step` for capacity searches. ([#2167](#))
- A static musl build for minimal container images. ([#2198](#))
- Live terminal dashboard, enabled with `--watch`. ([#2204](#))

Fixed

- Coordinated omission no longer hides stalls longer than the sampling interval. ([#2091](#))
- HTTP/2 connections are reused across scenario steps instead of per step. ([#2118](#))
- Reports keep their percentile order when a run is interrupted with `Ctrl-C`. ([#2150](#))

Deprecated

- The `--duration` flag; set `duration` in the scenario file instead. Removal is planned for 4.0. ([#2215](#))
- The Graphite exporter, in favour of OpenTelemetry. ([#2144](#))

Known issues

- The live dashboard flickers in terminals narrower than 80 columns. ([#2240](#))
- `kestrel migrate` drops YAML comments; review converted files before committing them. ([#2236](#))

Compatibility

Kestrel 3.0 raises the minimum versions on most platforms. Check Table 1 before you upgrade agents on older hosts.

Table 1. Supported platforms and minimum versions

Platform	2.9	3.0	Notes
Linux x86-64 (glibc)	2.28	2.31	Ubuntu 20.04, RHEL 9 and later
Linux arm64 (glibc)	2.28	2.31	Graviton and Ampere tested
Linux x86-64 (musl)		1.2	New static build for containers
macOS arm64	12	13	Ventura and later
macOS x86-64	12		Removed; use 2.9 LTS
Windows x86-64	10	10	Server 2019 and later
Kubernetes operator	1.26	1.28	Helm chart 5.x

2.9 LTS receives security fixes until 30 September 2027.

Upgrading from 2.x

1. Upgrade to 2.9.2 first and run your suite once; it warns about every deprecated flag.
2. Convert scenarios with `kestrel migrate scenarios/ --to toml --write`.
3. Install 3.0 and run `kestrel check` on every converted scenario.
4. Run once to create `kestrel.lock`, then commit it beside the scenarios.

Table 2. Flags and settings changed in 3.0

2.x	3.0	Notes
<code>--rps 500</code>	<code>--rate 500/s</code>	Units are required; per-minute rates use <code>/m</code> .
<code>--duration 10m</code>	<code>duration = "10m"</code>	Set in the scenario file; the flag still works until 4.0.
<code>scenario.yaml</code>	<code>scenario.toml</code>	Convert with <code>kestrel migrate</code> .
<code>--export statsd</code>	<code>export = "statsd"</code>	OpenTelemetry is the new default exporter.
<code>--graphite-host</code>		Removed with the Graphite exporter.
<code>--no-seed</code>		Removed: every run is seeded from <code>kestrel.lock</code> .

Questions are welcome in [the discussion forum](#).

2.9.2 · 12 August 2026

- Warns when a scenario uses a flag removed in 3.0. ([#2176](#))
- Fixed a crash when a target returned an empty `Content-Type` header. ([#2171](#))

2.9.1 · 29 July 2026

- Security: updated the bundled TLS library to address CVE-2026-31337. ([#2160](#))
- Histogram buckets above 60 s are no longer merged. ([#2158](#))

Thanks

Kestrel 3.0 includes work from 64 contributors, 23 of them new. Special thanks to **Amara Okafor** for the arrival models, **Tomás Ribeiro** for the TOML migration, and **Yuki Tanaka** for the coordinated-omission fix. Kestrel is released under the MPL-2.0.